Project Overview

The goal of this project was to explore **passive RF monitoring using software-defined radio (SDR)**. I used a HackRF One with a PortaPack H2 to receive ADS-B transmissions from aircraft in the local area. I then decoded the received data and used publicly available sources to correlate aircraft identifiers with additional information.

The project demonstrates how publicly broadcast RF signals can be collected, decoded, and combined with **open-source intelligence (OSINT)** to investigate information about aircraft without relying solely on existing aircraft-tracking websites.

Why ADS-B?

I chose ADS-B (**Automatic Dependent Surveillance–Broadcast**) because aircraft continuously broadcast information that can be received by compatible equipment on the ground.

For this project, I wanted to explore what could be learned by **directly receiving and decoding these broadcasts rather than simply viewing aircraft information through an existing tracking website**.

This creates a simple RF-to-OSINT workflow:

Aircraft

↓

ADS-B Broadcast

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SDR Reception

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ADS-B Decoding

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Aircraft Identifier

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Public Information

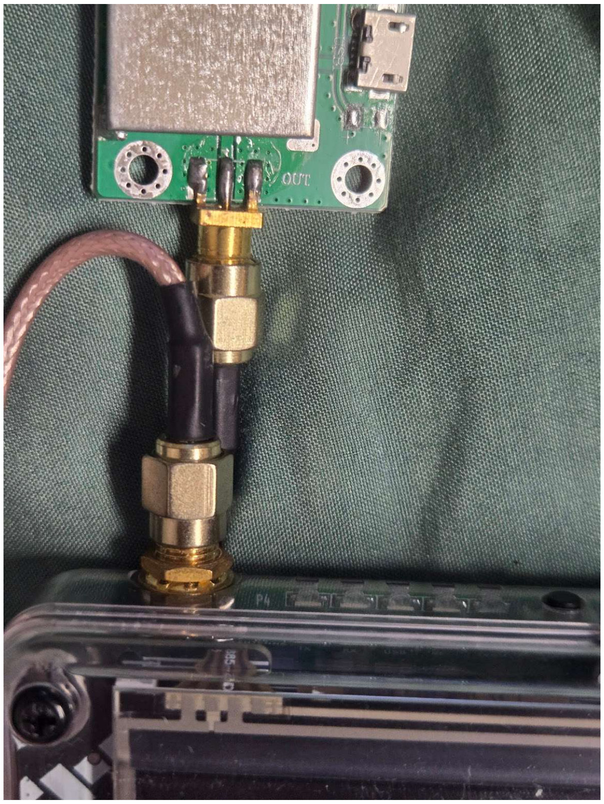
↓

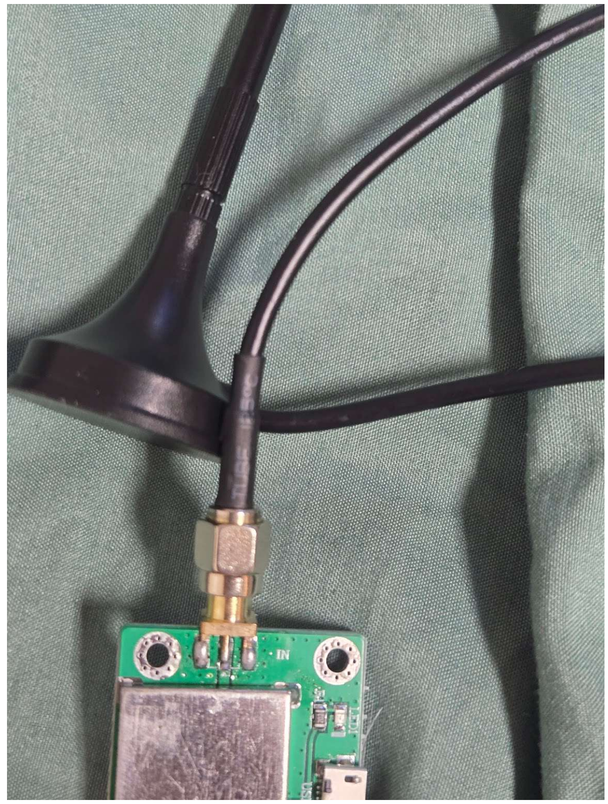
OSINT Correlation

Portapack h2 with a 5 watt amp and a 700mhz-2700mhz wideband whip antenna

The RF amplifier was connected between the SDR and antenna using the appropriate RF connections. The amplifier also requires a USB Micro-B power connection to operate.

Important: The orientation of the amplifier connections matters. The connection labeled **OUT** was connected toward the SDR, while **IN** was connected toward the antenna.





Out goes on the side with the h2 portapack and in goes to the antenna.

*The amp need a usb micro power cord to work

Because I was conducting the experiment from a dormitory and the building was constructed primarily of concrete, I positioned the whip antenna toward the window. RF signals can be affected by physical obstructions and building materials, which can reduce reception quality.

Additionally, operating in a campus environment means there may be numerous nearby RF sources and potential interference. I therefore experimented with antenna positioning and receiver gain to improve ADS-B reception.

LNA Gain

The **Low-Noise Amplifier (LNA)** controls the amount of gain applied early in the RF receive chain. Increasing LNA gain can make weaker ADS-B signals easier to detect, but excessive gain can also amplify unwanted noise and interference.

VGA Gain

The **Variable Gain Amplifier (VGA)** provides additional adjustable gain further along the receiver chain. I adjusted the VGA gain to find a balance between receiving weaker aircraft signals and avoiding excessive noise.

Amplifier Setting

The receiver also provides an amplifier setting that affects the overall signal level entering the processing chain. This was adjusted along with the LNA and VGA settings to optimize reception.

> The optimal gain settings are environment-dependent and may vary based on antenna placement, nearby RF interference, and signal strength.Notice the blue and red that's a gain/noise indicator. If there is no red it may be too weak to pick up anything but if the red is all the way across you could overload the receiver or increase the effects of interference

For example, during testing I found that:

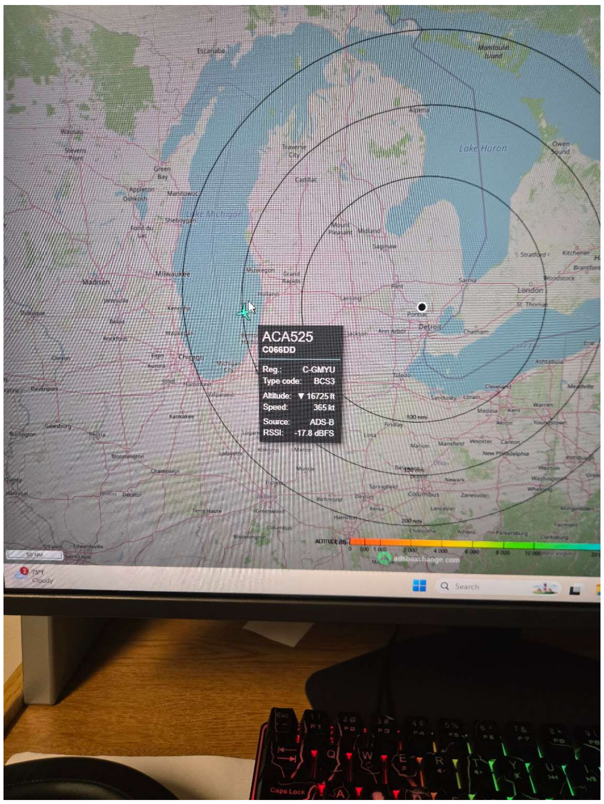
LNA: 24
VGA: 32
AMP: 1

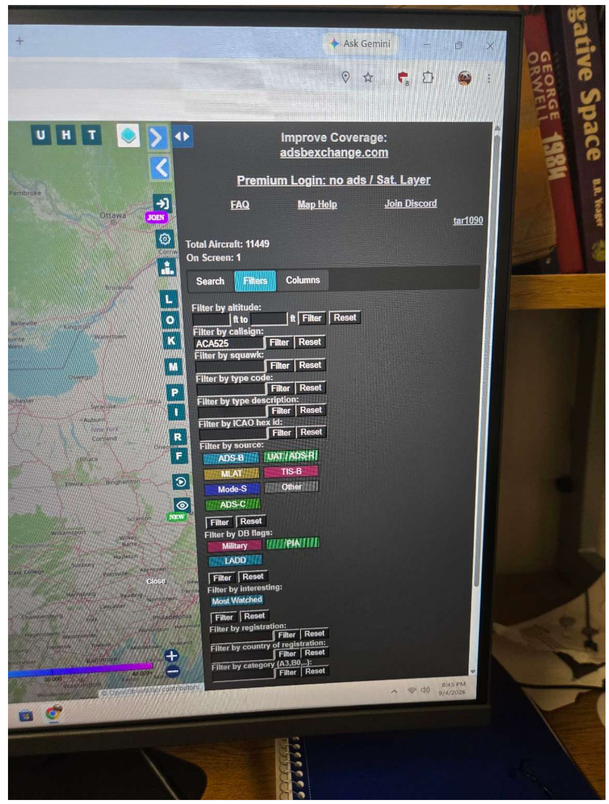
Produced the best aircraft detection for my environment.

Aircraft Identification & OSINT Correlation

The ADS-B receiver displays aircraft identifiers associated with the messages it successfully decodes. I used the ICAO24 identifier and callsign information displayed by the receiver as starting points for further investigation.

For this project, I used **ADS-B Exchange** to correlate the identifier received by my SDR with publicly available aircraft information.





The database provided additional information about the aircraft and demonstrated how an identifier obtained directly through RF reception can be correlated with publicly available information.

Why Not Just Use an Aircraft-Tracking Website?

A natural question is: **Why use an SDR to receive ADS-B when websites already provide aircraft-tracking information?**

Websites such as aircraft-tracking services make aircraft information convenient to access, but they abstract away the underlying RF and decoding process. The purpose of this project was not simply to find information about an aircraft. Instead, the goal was to understand the process of **collecting the underlying broadcast data and performing the analysis independently**.

Future Development

If I continued developing this project, I could automate the process of saving decoded ADS-B data locally. This would allow me to build my own aircraft-tracking application that could store, process, and visualize the data collected by my SDR. Instead of relying on an existing tracking website, I could develop my own system for displaying aircraft, recording historical observations, and performing additional analysis on the collected data.